VIMALRAJ D

AI/ML Engineer | Machine Learning Engineer | GenAI & LLM Engineer | LLM Engineer | GenAI Specialist
Backend ML Engineer | MLOps Engineer | NLP & Computer Vision

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of hands-on experience building production-ready Generative AI (GenAI), NLP, and Computer Vision systems across two internships. Skilled in LLMs, Retrieval-Augmented Generation (RAG), LangChain, and AI-driven backend services using Python, FastAPI, PyTorch, and MongoDB Vector Search. Built and deployed 3+ production-style RAG/LLM and Computer Vision pipelines, cutting manual processing time by 30-40% and containerizing services with Docker and GitHub Actions CI/CD. Passionate about designing scalable, efficient AI/ML solutions with clean architecture and strong performance optimization.

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning, Deep Learning, Generative AI (GenAI), Agentic AI, Large Language Models (LLMs), NLP, Computer Vision, CNN, RNN, LSTM, GAN, LoRA, PEFT, Model Training, Fine-Tuning, Model Optimization

LLM & GenAI Systems: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Search, Vector Databases, Conversational AI, AI Agents

Frameworks & Libraries: Hugging Face Transformers, PyTorch, TensorFlow-compatible workflows, Scikit-Learn, OpenCV, YOLO, NumPy, Pandas

Backend, APIs & Databases: FastAPI, REST API Development, Flask, Streamlit, MongoDB (Vector Search), SQL, NoSQL

Programming Languages: Python, SQL, JavaScript, HTML, CSS

Data Handling: Data Analysis, Data Processing, OCR, PDF & Document Processing, Data Pipelines

Cloud, DevOps & MLOps: Docker, CI/CD (GitHub Actions), AWS, Git, GitHub, Power BI

CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Software Engineering

WORK EXPERIENCE

Product Engineer Intern (AI/ML) - ConverSight AI Apr 2026 - Present

- Contributed to Athena, a Generative AI-powered conversational AI platform, improving response reliability and UX for enterprise data queries.
- Cut data processing latency by 20% by developing and integrating 5+ Python backend APIs.
- Built ML notebooks for data exploration, model experimentation, and insight generation, accelerating iteration speed for the AI/ML team.
- Executed platform version migrations across the AI system, ensuring backward compatibility and zero-downtime upgrades.

Selected Technologies: Python, ML, Generative AI, Conversational AI, FastAPI, REST APIs, Docker, CI/CD (GitHub Actions), Git

AI/ML Developer Intern - CubeAI Solutions Mar 2025 - Jun 2025

- Developed AI-driven automation for construction procurement and on-site operations, reducing manual processing time by 30-40%.
- Built a LangChain-based LLM assistant using RAG that queried 1,000+ documents in under 10 seconds.
- Contributed to a YOLO-based Computer Vision material detection system and analytics dashboards.

Selected Technologies: Python, FastAPI, LangChain, MongoDB, LLMs, RAG, YOLO, Computer Vision, REST APIs, Git

PROJECTS

Finance Agent AI (GitHub: github.com/Vimalraj-D/Stock_Market_Analyser) Aug 2025 - Oct 2025

- Built an AI-powered stock analysis system covering 50+ tickers with sub-2-second response time using LLM-driven data parsing.
- Developed a FastAPI backend with interactive, real-time dashboards for market trend visualization and portfolio tracking.
- Implemented intelligent data parsing and structured AI-generated responses to support reliable financial decision-making.
- Optimized API response handling and caching, reducing average query latency and improving overall system throughput.

Tech Stack: Python, FastAPI, LLMs, Chart.js, HTML, CSS, JavaScript

AI Food Analyser - Computer Vision (GitHub: github.com/Vimalraj-D/AI_Food_Analyser) Feb 2025 - Mar 2025

- Built an AI-based food recognition system achieving 88% classification accuracy across 50+ food categories using YOLO and Computer Vision techniques.
- Generated automated nutritional analysis and personalized meal recommendations directly from food images.
- Integrated ingredient-based recipe suggestions using Machine Learning models for smart cooking guidance.
- Designed an end-to-end inference pipeline covering image preprocessing, detection, and results rendering for fast, real-time predictions.

Tech Stack: Python, OpenCV, YOLO, Computer Vision, Machine Learning Models

Book Recommendation System (GitHub: github.com/Vimalraj-D/Book-Recommendation) Jul 2024 - Aug 2024

- Improved recommendation accuracy by 15% over baseline on 10,000+ ratings using a collaborative filtering Machine Learning engine.
- Designed and deployed a Flask-based web application with a responsive UI for personalized book recommendations.
- Engineered data preprocessing and similarity-scoring pipelines to handle sparse rating matrices efficiently.

Tech Stack: Python, Scikit-Learn, Pandas, Flask, HTML, CSS, JavaScript

CERTIFICATIONS

- MongoDB Skill Badge - Building RAG Apps Using MongoDB (Jan 2026)
- Coursera (AWS x DeepLearning.AI): Generative AI with Large Language Models (Oct 2025)
- Kaggle: Computer Vision Certification (May 2025)
- NVIDIA DLI: Getting Started with AI on Jetson Nano (Jan 2025)
- NPTEL: Introduction to Programming in C, Elite + Silver (Apr 2024)
- LinkedIn Learning: Learning Data Science - Understanding the Basics (Apr 2024)

EDUCATION

B.E. Computer Science and Engineering (AI & ML)

Dr. Mahalingam College of Engineering and Technology | 2023 - 2027 | CGPA: 8.4

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- Winner - Smartathon 2024: AI-based Innovation Challenge
- Technofete 2025: 3x First Prizes & 1x Second Prize in AI and Automation events
- Winner - Web Designing Contest (Hilaricas 2024)
- Active member - Youth Red Cross (YRC); participated in blood donation camps and community drives.